

Laplace Transform of $t^{-1/2}$ and Bromwich Contour

Source: J. E. Santos, Cambridge Part IB Complex Methods, Example Sheet 3, Question 12.

Using the equality $\int_0^\infty e^{-x^2} dx = \frac{1}{2}\sqrt{\pi}$, find the Laplace transform of $f(t) = t^{-1/2}$. By integrating around a Bromwich keyhole contour, verify the inversion formula for $f(t)$. What is the Laplace transform of $t^{1/2}$?

Solution. The integral can be evaluated easily by a change of variable $\tau^2 = t$:

$$\hat{F}(s) = \mathcal{L}[f(t)] = \int_0^\infty dt e^{-st} t^{-1/2} = \int_0^\infty d\tau 2e^{-s\tau^2} = \sqrt{\frac{\pi}{s}}. \quad (1.1)$$

Now, we shall verify the inverse Laplace transform of $\hat{F}(s)$. The function $\hat{F}(s)$ has a principal branch cut from $-\infty$ to 0. To see that, write $s = re^{i\theta}$, and we find

$$s^{-1/2} = r^{-1/2} e^{-i\theta/2} = r^{-1/2} e^{-i \arg s / 2} \quad \text{with} \quad \arg s \in (-\pi, \pi). \quad (1.2)$$

Therefore, just above the branch cut, we have $s^{-1/2} = -ir^{-1/2}$; just below the branch cut, we have $s^{-1/2} = ir^{-1/2}$.

The inverse Laplace transform is given by

$$f(t) = \frac{\sqrt{\pi}}{2\pi i} \int_{\mathcal{C}_\gamma} ds e^{st} s^{-1/2} = \frac{\sqrt{\pi}}{2\pi i} \int_{\gamma-\infty}^{\gamma+i\infty} ds e^{st} s^{-1/2} \quad (1.3)$$

with $\gamma \rightarrow 0^+$. We can choose to integrate the Bromwich keyhole contour, $\mathcal{C}_B = \Gamma_+ \cup \Gamma_- \cup \mathcal{C}_\epsilon \cup \mathcal{C}_\gamma \cup \mathcal{C}_+ \cup \mathcal{C}_-$, where

$$\Gamma_\pm = \lim_{\epsilon \rightarrow 0} \{z \in (\pm i\epsilon - \infty, \pm i\epsilon]\}, \quad \mathcal{C}_\epsilon = \lim_{\epsilon \rightarrow 0^+} \{z = \epsilon e^{i\theta} | \theta \in [-\pi/2, \pi/2]\}, \quad (1.4)$$

and the large arcs are given by

$$\mathcal{C}_\pm = \lim_{R \rightarrow \infty} \{z = Re^{i\theta} | \theta \in [\pm \frac{\pi}{2}, \pm \pi]\}. \quad (1.5)$$

Since there are no poles enclosed by this contour, we have

$$\oint_{\mathcal{C}_B} ds e^{st} s^{-1/2} = 0. \quad (1.6)$$

It is easy to see that the integral along $\mathcal{C}_\pm$ and $\mathcal{C}_\epsilon$ vanishes. The integral along $\Gamma_\pm$ is given by

$$I_{\pm} = \int_{\Gamma_{\pm}} ds e^{st} s^{-1/2} = \pm \int_{-\infty}^0 dr r^{-1/2} e^{rt} = \mp i \int_0^{\infty} dr r^{-1/2} e^{-rt} = \mp i \sqrt{\frac{\pi}{t}}. \quad (1.7)$$

Thus, we find (notice the direction of the contour integration)

$$2\sqrt{\pi}i f(t) + I_+ - I_- = 0 \implies f(t) = t^{-1/2}. \quad (1.8)$$